

Multimedia Data Management M

Second cycle degree programme (LM) in Computer Engineering
University of Bologna

Exercises

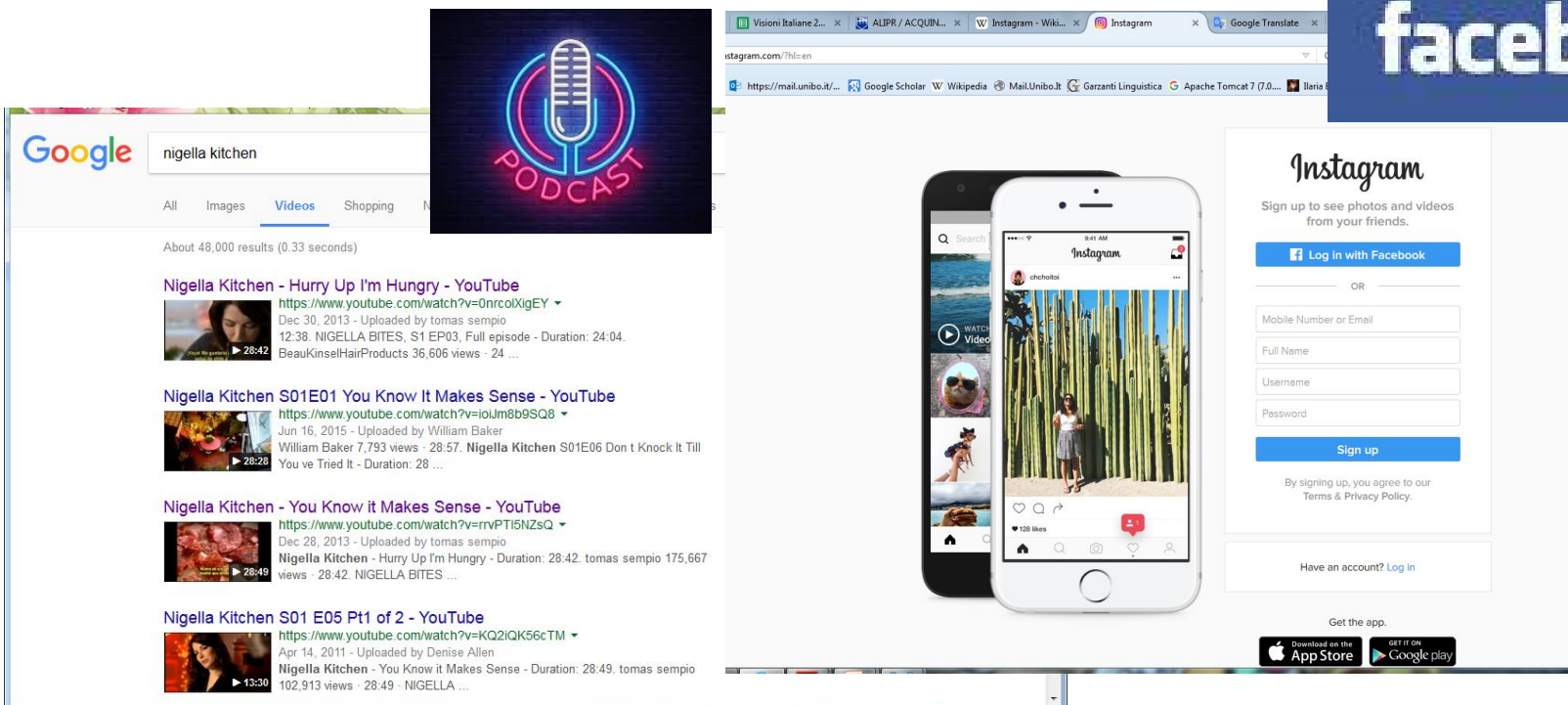
Home page: <http://www-db.disi.unibo.it/courses/MDM/>

Electronic version: 0.03.Exercises.pdf

Electronic version: 0.03.Exercises-2p.pdf

Exercises: working modality

- You may work alone or in team (max. 3/4 students)
- In both cases, before starting working, please, fill the form “**StudentsFreeExercise.doc**” (ref. OneDrive folder “MDM”) providing all requested information
 - If you decide to work in team, the reference name for the .ppt file is the one of the team “leader”



Exercise 1.A

- Make a note of all the different media and combinations of media you are exposed to in a single day
- Figurate some concrete examples of advanced **MM data applications** or **services** (like the ones just illustrated) by separately describing, in **natural language**, relevant **structured, semi-structured, and unstructured data representations** involved

Archivio Storico Fiat



- Trimotore Fiat G212
- Data: 1947
- Collezione: Tema di cultura industriale
- Tipologia: Immagine

Aereo, Motore, Ali



What else? ...



Exercises 1.A: students to do list

- Starting from the **media notes** and the **description of your advanced MM data applications**, reshape the description in natural language following the schema:
 - Advanced MM data application **proposal**;
 - Involved **use cases** (at least 3);
 - **Involved MM data**:
 - Textual documents, images, audio, videos, etc.
- Prepare an electronic version of your proposals
 - .ppt file
- Upload it into the **OneDrive folder “MDM”**, using the link provided by the teacher, and following the instruction:
 - **File name: “Secondname.Firstname.ESE1.A.ppt”**

Exercise 1.B

- Starting from **descriptions in natural language** of relevant **structured, semi-structured, and unstructured data representations** selected for your examples of Exercise 1.A, **model the data** according to:
 - **relational model** (for **structured data**), and
 - **XML model** (for **semi-structured data**)
- Provide a **definition** (in natural language) **accurate as much as possible of the characteristics** (i.e., **low-level features**) you chose for **describing the “content”** of involved MM data (**unstructured data representation**)

Exercises 1.B: students to do list

- Starting from the solution you proposed in Exercises 1.A (i.e., the file “**Secondname.Firstname.ESE1.A.ppt**”), enrich the description of your MM data applications following the examples proposed during lectures
- Upload it into the **OneDrive folder “MDM”**, using the link provided by the teacher, and following the instruction:
 - File name: “**Secondname.Firstname.ESE1.AB.ppt**”

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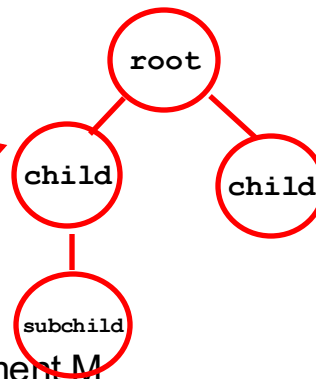


- Trimotore Fiat G212
- Data: 1947
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- Tipologia: Immagine



Descrizione low-level features per dati unstructured

- Distribuzione colore dell'immagine
- Forma degli oggetti nell'immagine
- ...



Exercise 1.C

- Starting from Examples proposed in Exercises 1.A/1.B, and by focusing on the **textual documents** only
- Draw
 - the **term-document matrix** and
 - the **inverted index**representations for Boolean and VS retrieval models
- Identify some relevant **queries**
- Plot the two separate graphs of
 - **Average Precision vs. Nret documents** and
 - **Average Recall vs. Nret documents**

Exercises 1.C: students to do list

- Starting from the solution you proposed in Exercises 1.A/1.B (i.e., the file “**Secondname.Firstname.ESE1.AB.ppt**”), enrich the description of your MM Applications by adding
 - **visual examples** (i.e., **images**) of involved **relevant textual information**
 - provide a sketch of the **VS model-based inverted index** you built for solving your IR queries
- Upload it into the **OneDrive folder “MDM”** the updated electronic version of your work following the instruction:
 - **File name:** “**Secondname.Firstname.ESE1.ABC.ppt**”

Exercise 1.D

- Starting from the **definitions of the low-level features** you selected for describing the “**content**” of **unstructured data** representations involved in your MM applications, provide a **concrete representation/comparison modality** of them with visual examples
- Among features possibilities:
 - global features vs. local features (region-based approach) and/or
 - local salient points

E.g., “*global color distribution for an image*” (definition) vs. “*color histograms*” (representation) by using the “*weighted Euclidean distance*” as similarity measure (comparison modality)

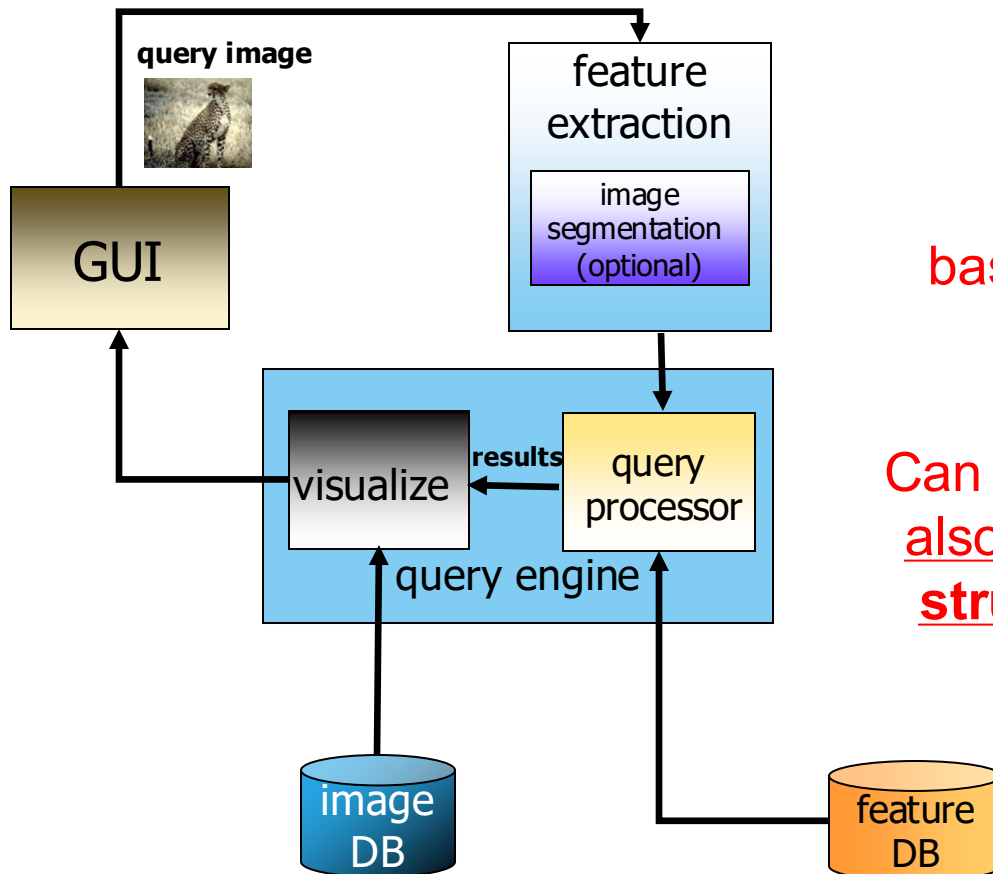
Exercise 1.D: students to do list

- Starting from the solution you proposed in Exercise 1.C (i.e., the file “**Secondname.Firstname.ESE1.ABC.ppt**”), enrich the description of your MM Applications by adding
 - **concrete representation/comparison modality** of involved **low-level features** enriched by means of **visual examples** (i.e., thumbnails)
- Upload it into the **OneDrive folder “MDM”** the updated electronic version of your work following the instruction:
 - File name: “**Secondname.Firstname.ESE1.ABCD.ppt**”

In doing the exercise, let's keep in mind the final goal:
“Retrieve/Recommend relevant/useful MM content”! 😊

Exercise 1.E: from modeling to designing

- Let's map your MM data applications/services (with all unstructured data representations, you have modeled) to the reference software architecture for any general MM data application/service



You can map your specific MM data application to the basic software modules, right? 😊

Can you extend your architecture by also integrating descriptors of your structured and semi-structured data representations?

Exercise 1.E: students to do list

- Starting from the solution you proposed in Exercise 1.D (i.e., the file “**Secondname.Firstname.ESE1.ABCD.ppt**”), add slides for illustrating the design of the “tailored” software architecture of your MM data application/service
- Upload it into the **OneDrive folder “MDM”** the updated electronic version of your work following the instruction:
 - File name: “**Secondname.Firstname.ESE1.ABCDE.ppt**”

Exercise 1.F: from design to implementation (1/2)

- Summarizing, the **Multimedia Data Management** course presents **classic MM pipelines** based on **explicit feature extraction** and **exact similarity search** as foundational baselines, and then **bridges them to modern multimodal embedding approaches** (e.g., contrastive image–text models) to study how **representation choices affect indexing strategies, retrieval quality, scalability, and latency-accuracy trade-offs** in **contemporary MM data applications/systems**
- We conclude our **practical exercise sessions** showing the **implementation of two retrieval pipelines** over the same multimodal dataset to make explicit the evolution from foundational baselines (mid-2010s) MM data systems to current practices

Exercise 1.F: from design to implementation (2/2)

- The **first pipeline** follows a **classic design** based on structured metadata, lexical matching, and exact similarity computation, emphasizing **interpretability** and **well-defined guarantees**
- The **second pipeline** replaces manual feature engineering with a **modern multimodal embedding model** and supports scalable retrieval using approximate nearest neighbor indexing while **keeping the underlying data management principles** (modeling, indexing, and evaluation) **constant**